# GERMÁN KÜBER

## Staff AI Engineer — Production LLM Agents, Agent Evaluation & Enterprise Solution Delivery

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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## PROFESSIONAL SUMMARY

Staff-level AI engineer with 15+ years building and delivering production systems, and recent focus on shipping production AI agents, agentic workflows, multi-agent systems, and real-time AI. Currently AI Architect at Nera, owning the company-wide knowledge and code-intelligence layer (a Neo4j + pgvector ontology over Notion, Jira and GitHub, and a 66-repo code graph served through MCP) and the multi-agent LangGraph tooling that takes product and engineering from problem to spec, with evals as a release gate. Client-facing throughout: previously a forward-deployed independent AI consultant shipping production agentic workflows (LangGraph, Microsoft Agent Framework, A2A protocol), RAG platforms and knowledge-graph agent memory for an enterprise client, and before that the visible technical lead across 6+ enterprise clients in banking, telecom, healthcare and education, running discovery with business stakeholders, architecting the solution, and owning delivery on-site through to production. Shipped and operated multi-provider LLM orchestration spanning **OpenAI, Anthropic, Google Gemini**, plus managed cloud services (AWS Bedrock, Azure OpenAI Service, Google Vertex AI), with dynamic model routing for cost and performance optimization. Hands-on with MCP servers and clients, function calling, structured outputs, and ReAct-style agent patterns. Built production RAG pipelines (Pinecone, Chroma, pgvector, Redis) with semantic chunking, reranking and hybrid retrieval, and instrumented a full LLM evaluation and observability stack (LangSmith, RAGAS, DeepEval, Langfuse, OpenTelemetry GenAI conventions), including eval safety and red-teaming frameworks that measurably reduced hallucination rate. Real-time voice AI with OpenAI Realtime API, Deepgram and ElevenLabs at sub-200ms TTFT. Cut combined LLM and cloud spend by 45% through semantic caching, prompt compression, and dynamic model selection. Polyglot across **Python, TypeScript / Node.js, React, Go, .NET and Rust**. Comfortable operating end-to-end in ambiguous, fast-moving environments — from customer discovery to production deployment and continuous iteration. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

## CORE EXPERTISE

**Solution Delivery & Enterprise AI Deployment:** Forward-deployed / customer-facing / client-facing engineering, solution architecture, enterprise AI deployment, requirements discovery with business stakeholders, pilot-to-production ownership, technical due diligence, stakeholder communication across technical and non-technical audiences, technical pre-sales support, on-site delivery in regulated industries (banking, telecom, healthcare, education, iGaming)

**AI / LLM Systems:** Multi-provider LLM orchestration (OpenAI, Anthropic Claude, Google Gemini, Llama), AWS Bedrock, Azure OpenAI Service, Google Vertex AI, dynamic model routing, prompt engineering, context engineering, structured outputs (JSON schema validation), function calling / tool calling, cost optimization (semantic caching, prompt compression, model selection), fine-tuning experimentation (LoRA / QLoRA)

**Multi-Agent & Agent Engineering:** production AI agents, agent development life cycle (pilot to deployment to continuous iteration), agentic workflows, agent frameworks and tooling, LangChain v1, LangGraph, Microsoft Agent Framework (MAF), multi-agent orchestration (supervisor pattern, conditional routing), A2A (Agent2Agent) protocol, Agent Cards, agent capability registries, ReAct pattern, state machines, tool calling, function calling, human-in-the-loop, durable execution, agent semantic memory, agent tuning and iteration from production traces, MCP (Model Context Protocol) — server and client implementation

**Knowledge Graphs & Agent Memory:** Neo4j, ontology design (classes, invariants), pgvector semantic indexes, Cognee, Graphiti, company-brain knowledge graphs, code architecture graphs for agent-queryable system structure (static analysis + OpenTelemetry runtime calls), semantic memory layers, context engineering, enterprise knowledge structuring for LLM consumption

**Vector Databases & RAG:** Pinecone, Chroma, pgvector, Redis (RediSearch), PostgreSQL, semantic search, hybrid retrieval, chunking strategies, semantic chunking, embeddings, embedding models, reranking (cross-encoders), retrieval recall@k, production-grade RAG pipelines

**Real-Time AI & Voice:** OpenAI Realtime API, Deepgram (STT), ElevenLabs (TTS), SignalR, WebSocket streaming, real-time audio pipelines, sub-200ms TTFT (Time To First Token), voice activity detection, barge-in handling

**AI Safety, Guardrails & Red-Teaming:** Systematic eval safety frameworks, red-teaming pipelines, hallucination mitigation, output guardrails, content filtering, JSON schema validation for structured outputs, adversarial testing

**LLM Evaluation & Observability:** LangSmith, RAGAS, DeepEval, Langfuse, OpenTelemetry GenAI semantic conventions, eval frameworks (faithfulness, correctness, answer relevancy), LLM-as-judge, record-and-replay harnesses, evals as release gates, agent task completion rate, hallucination rate measurement, trace analysis, regression testing, production monitoring

**Programming Languages:** Python (FastAPI, async), TypeScript / Node.js, React, Go, JavaScript, C# / .NET, Rust, Cython, SQL

**Cloud & AI Infrastructure:** AWS (Bedrock, SageMaker, Lambda, S3, ECS), Azure (OpenAI Service, AI Search), Google Cloud (Vertex AI), Docker, Kubernetes, FastAPI, CI / CD, Prometheus, observability stacks

**ML / Inference:** PyTorch, Ollama (local inference), Hugging Face

**Quantum Computing (learning):** Qiskit, Q#

**Distributed Systems:** Async runtimes, low-latency networking, fault tolerance, microservices, event-driven architectures, high-throughput pipelines, real-time signal processing

**Programming Paradigms & Design:** Object-Oriented Programming (OOP), Functional Programming, design patterns, architecture patterns, data structures, algorithm design, SOLID principles, clean code

**Engineering Practices & Leadership:** Test-Driven Development (TDD), pair programming, code review, mentoring, technical interviewing, ADRs, design reviews, CI / CD, DevOps practices, Infrastructure as Code (Terraform), technical roadmap, on-call and incident response

## PROFESSIONAL EXPERIENCE

### AI Architect

*Nera — Feb 2026 – Present (full-time) — Remote*

- Own Nera's AI architecture: the knowledge and code-intelligence layer that lets the whole company query the business in natural language, and the agentic tooling product and engineering use to go from problem to spec.
- Designed and built a company-wide ontology (28 classes, 26 invariants) that ingests Notion, Jira and GitHub into a Neo4j knowledge graph plus a pgvector semantic index, giving every area end-to-end traceability from problem to PR and natural-language Q&A over MCP.
- Built a code graph of the entire microservice fleet (66 repos, 77k nodes, 210k relationships across Java, TypeScript and Go), enriched with runtime calls from OpenTelemetry and served through a 32-tool MCP server that engineers use to design features and product uses to write PRDs.
- Built an AI product-engineering platform where multi-agent LangGraph flows take a problem through PAD, PRD, epic and tech spec, grounded in the code graph and business knowledge.
- Made evals a release gate: record-and-replay harnesses, a 100-question benchmark for the knowledge graph, fidelity audits for the code graph and LLM-as-judge for generated PRDs.

- Automated user-research capture with a pipeline that transcribes interviews and extracts verifiable quotes and insights into Notion with human approval, replacing a manual tracker.
- Built AI tooling for product and design: a Claude plugin for research, interview guides and synthesis, and a Figma plugin that audits screens against the design system.
- Drove Spec-Driven Development (SDD) across engineering with a shared framework (specs, TDD, review gates) that standardized how features are built and raised quality team-wide.
- Organized Nera's internal training space and gave talks on engineering cognition, LLM architecture beyond the prompt, Vertical Slice Architecture and HTTP internals.

**Independent AI Consultant — Agent Engineering & Enterprise AI Delivery**

*Self-employed (autónomo, Spain) — Mar 2026 – Jul 2026 (parallel to Nera) — Remote, client under NDA*

- Forward-deployed AI consultant embedded with an enterprise engineering organization: worked directly with the customer to run discovery with business and technical stakeholders, architected the agent solution, and owned delivery from pilot through production and continuous iteration.
- Designed and shipped production agentic workflows on LangChain, LangGraph and Microsoft Agent Framework — testing each graph, running eval frameworks and eval suites against it, and monitoring it live in production to tune behavior from real traces.
- Built multi-agent architectures on Google's A2A (Agent2Agent) protocol, exposing agent capabilities as Agent Cards and designing a central capability registry so functionality is discoverable and reusable across agents.
- Owned the RAG platform end-to-end for internal company documentation: retrieval optimization, chunking and context engineering, plus regression test suites that gate every change to a pipeline before it reaches production.
- Led context engineering as a discipline — helping the organization structure and bucket its internal knowledge so it is actually consumable by LLMs, and advising on which data matters for long-term decision-making.
- Cut inference cost and latency through semantic caching, agent-level semantic memory, and dynamic model routing across providers.
- Fine-tuned models with LoRA on internal company documents and deployed them through Azure AI Foundry to improve agent accuracy on domain-specific tasks.
- Built "company brain" knowledge graphs with Cognee and Graphiti over internal repositories, including specialized code graphs that map microservice architecture so agents can query real system structure when making decisions.
- Designed an end-to-end agentic pipeline running from company discovery through to production deployment with no human intervention, with targeted human-in-the-loop checkpoints placed to guarantee quality.
- Introduced a Spec-Driven Development (SDD) framework across the delivery process — versioning, testing and distributing reusable agent skills, and measuring the improvement they produce.

**Lead AI Architect & Co-Founder**

*KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain*

- Shipped multi-provider LLM orchestration integrating OpenAI, Anthropic Claude, and Google Gemini through both direct APIs and managed cloud services (AWS Bedrock, Azure OpenAI Service, Google Vertex AI), with dynamic model routing to pick the best model per task and cut inference spend.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms TTFT on voice pipelines (OpenAI Realtime API).
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, Whisper, Play.ht) on latency, cost and quality — reaching 90ms latency and nearly 95% word accuracy.
- Built and shipped production AI agents on LangGraph — an adaptive simulation engine with multi-agent architecture including state machines, conditional routing, ReAct-style reasoning, multi-step branching logic, and RAG-powered personalization — owning the full agent life cycle from pilot through deployment and continuous tuning against production traces.
- Drove agent quality with an eval-first workflow: built eval frameworks and eval suites in LangSmith, Langfuse, RAGAS and DeepEval, then iterated agent prompts, tools and routing off real production traces and regression scores rather than anecdote.
- Shipped production RAG pipelines across Pinecone, Chroma, pgvector, and Redis (RediSearch), with semantic search, hybrid retrieval, reranking via cross-encoders, and chunking strategies tuned per use case; implemented function calling and structured outputs (JSON schema validation) for reliable agent-tool integration.
- Architected real-time voice AI pipelines (OpenAI Realtime API, Deepgram, ElevenLabs, SignalR) at sub-200ms TTFT across concurrent sessions, and implemented MCP servers and clients to standardize tool access across LLM workflows.
- Designed and ran systematic eval safety and red-teaming frameworks, significantly reducing hallucination rate across critical paths; also built a Zero-Knowledge proof system (Noir, Barretenberg, Garaga on StarkNet) for verifiable expert credentials.
- Instrumented LLM evaluation and observability stack using LangSmith, RAGAS, DeepEval, Langfuse, and OpenTelemetry GenAI semantic conventions for regression testing, faithfulness / correctness scoring, and production monitoring.
- Owned technical strategy, architectural decisions, and delivery as the senior-most engineer; led platform development end-to-end on a polyglot stack (Node.js / TypeScript, Python, .NET, Rust, Kubernetes, AWS). Hired and led a distributed team of up to 3 engineers.

**Fractional CTO (Part-time, hands-on)**

*Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote*

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built and deployed an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, working directly with the client to frame the business problem — raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Benchmarked LLM usage and optimized cloud infrastructure through semantic caching, prompt compression, and dynamic model selection across providers, cutting combined API and cloud spend by 45%.
- Designed a reusable base architecture on LangChain and LangGraph with monitoring via Langfuse and LangSmith and prompt versioning; migrated the real-time LLM service from OpenAI to Azure AI Foundry for compliance and data retention, cutting latency and improving scalability.
- Introduced Spec-Driven Development (SDD) and AI-driven development workflows with autonomous agents automating code review, testing, and deployment.
- Shipped AI-powered features in Node.js / TypeScript and Python (FastAPI, LangChain), instrumented the team's first LLM evaluation and monitoring with LangSmith and Langfuse, and ran weekly public-speaking workshops to sharpen team communication.

**Lead Software Engineer (Distributed Systems / Blockchain)**

*Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote*

- Lead engineer for low-latency distributed systems in Rust (tokio, async), Go, and Python / Cython; pipelines processed 10K+ opportunities per day from $50M+ in daily on-chain flow and contributed $100K+ in attributable annual P&L with execution under 10ms.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.

- Built high-throughput data pipelines and real-time signal processing across multi-chain ecosystems (EVM, Solana), with custom transaction delivery infrastructure achieving sub-slot inclusion on Solana (~400ms); reverse-engineered Geth and Reth to improve transaction simulation and gas estimation, contributing to the REVM community.
- Owned 24/7 production reliability and incident response, including on-call rotation, post-mortems, and observability ownership with Prometheus and custom Rust telemetry tracking 50+ real-time metrics.
- Ran technical due diligence and advisory engagements on 5+ projects that ended up getting funded — meeting external founders and engineering teams directly to assess architecture, risk, and feasibility, then presenting findings to investment stakeholders; acted as the CTO's technical reference on the low-level side.
- Acted as technical lead across the blockchain team, which grew from 4 to 10 engineers: code review, mentoring, technical interviewing, and architectural decisions.

### Senior Software Architect

*Paramo Technology — Oct 2020 – Sep 2021 — Remote*

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed and rolled out event-driven microservices in .NET (C#, ASP.NET Core) for real-time game state, wallet operations and regulatory reporting; migrated relational databases to MongoDB where it made sense.
- Implemented distributed tracing across all services (OpenTelemetry, Application Insights, Datadog); brought in Kubernetes to scale and built automated deploy pipelines.
- Led part of the hiring and designed the technical entry exam; created and delivered internal training on architecture patterns, testing discipline, and operational best practices.

### Software Architect

*CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)*

- Was the company's first employee; acted as the visible technical lead and primary point of contact for 6+ enterprise clients across banking, telecom, healthcare and education — running requirements discovery with business stakeholders, architecting the solution, and delivering on-site through to production.
- Owned solution architecture end-to-end across several enterprise product lines: translated ambiguous business problems into shippable systems, then stayed with each engagement through deployment and iteration.
- Designed Angular and React frontends with reusable component libraries and Node.js / TypeScript and .NET microservices backends using Domain-Driven Design; built a distributed system over geospatial data with PostGIS for a UK client.
- Delivered technical training and conference talks to partners and internal teams; communicated architecture and tradeoffs to both engineering and executive audiences.

### Software Architect / Team Lead — Earlier Roles

*Majestic Hotel Group (2014–2017) · Plus54 (2011–2014) · Acsys (2009–2011) · IBM (2008–2009) — Argentina*

- Full-stack Angular and .NET for hotel management platforms, first as Team Lead then architect; led the AngularJS to Angular 2 migration and redesigned the data model toward event-oriented architecture. Earlier: .NET web applications, internal tooling and mobile components end-to-end.

## KEY ACHIEVEMENTS

**Company-Wide Knowledge & Code Intelligence at Nera** — 28-class ontology over Notion, Jira and GitHub in Neo4j + pgvector, plus a 66-repo / 77k-node code graph served through a 32-tool MCP server, with a 100-question benchmark and LLM-as-judge evals gating every release.

**End-to-End Agentic Delivery Pipeline (Independent Consulting)** — Designed an agentic process running from company discovery to production deployment without human intervention, with targeted human-in-the-loop checkpoints for quality, backed by knowledge graphs (Cognee, Graphiti) and eval-gated regression testing.

**Production AI Agents at KliverAI** — Multi-agent system on LangGraph taken from pilot to production, with eval-driven iteration (LangSmith, Langfuse, RAGAS, DeepEval), multi-provider LLM orchestration across OpenAI, Anthropic, Gemini, AWS Bedrock, Azure OpenAI and Vertex AI, and sub-200ms voice latency.

**Enterprise Solution Delivery across 6+ Clients** — Visible technical lead and client-facing architect at CloudX for banking, telecom, healthcare and education clients, owning discovery, architecture and on-site delivery through to production.

**45% LLM Cost Reduction at Rocking Product** — Cut combined API and cloud spend through semantic caching, prompt compression, and dynamic model routing in production.

**Microsoft MVP, Developer Technologies** — 8 consecutive years (2017–present), recognized for technical content and community contributions.

**80+ International Technical Talks** — .NET Conf, DevFest, NetCoreConf, vOpen, Global Azure Bootcamp, and meetups across LATAM and Europe.

**Net-Baires Founder** — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.

**Polkadot Builder Program Graduate** — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).

**Microsoft Student Partner** — 5 years (2016–2021).

## EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft

Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft

Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)

Quantum Computing (ongoing self-study): quantum circuits and algorithms with Qiskit (Python) and Q# (Microsoft Quantum Development Kit).

Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

## OPEN SOURCE & COMMUNITY LEADERSHIP

**Net-Baires (.NET Community)** — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.

**Crypto Para Mortales** — Founder (2023–present), Spanish-language blockchain education conference. **Patrones Para Mortales** — Founder (2020–present), content brand on design patterns and architecture. **YouTube (youtube.com/@germankuber)** — deep-dives on Rust, blockchain, AI and architecture.

## SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams